The Complete Structural Mereology of a Neurosymbolic Search-and-Rescue Manifold: A Unified Axiomatic System Specification

System Architecture Invariant Manifest

October 23, 2026

1 Introduction and Architectural Core

We formalize the search-and-rescue plot for a central Fire-Station as an interlocking multi-graph Sequential Dynamic System (SDS) operating across distinct modal boundaries. Following Edward Zalta's Theory of Abstract Objects, the spatiotemporal execution plane maps to ordinary object exemplification (Fx), while the invariant control forests encode properties atemporally (xF).

2 Distributed Multi-Directory Agent Initializations

Following Barwise's structural Information Flow frameworks, multi-agent domain isolation requires true file-system sandboxing to prevent VRAM memory leakages. Each independent neuro-symbolic agent $a_k \in \mathcal{A}$ maintains its unique background logic constraints inside a dedicated disk directory path:

- `agent_0_master/kb.pl`: Core macro-systemic invariants.
- `agent_1_specialist/kb.pl`: Searle social reasoning world axioms.
- `agent_2_worker/kb.pl`: Kinematic search-and-rescue situation isomorphisms.
- `agent_3_observer/kb.pl`: Public base-rate diagnostic predicates.

3 The Eight Rasiowa Satisfaction Clauses for Multi-Agent Consensus

Let $\mathcal{M} = \langle A_t \rangle_{t \in T}$ be a T-reality over a branching poset lattice where $w \preceq t$. The consensus satisfaction relation $\mathcal{M} \models d_t(\phi)[v]$ under variable valuation v unrolls across eight rigorous algebraic operators:

1. $\mathcal{M} \models d_t(p_i(x_1, \ldots, x_m))[v] \iff A_t \models p_i(x_1, \ldots, x_m)[v]$
2. $\mathcal{M} \models d_t(\phi \vee \psi)[v] \iff \mathcal{M} \models d_t(\phi)[v]$ or $\mathcal{M} \models d_t(\psi)[v]$
3. $\mathcal{M} \models d_t(\phi \wedge \psi)[v] \iff \mathcal{M} \models d_t(\phi)[v]$ and $\mathcal{M} \models d_t(\psi)[v]$
4. $\mathcal{M} \models d_t(\phi \implies \psi)[v] \iff \forall s \preceq t, \mathcal{M} \models d_s(\phi)[v] \implies \mathcal{M} \models d_s(\psi)[v]$

5. $\mathcal{M} \models d_t(\neg\phi)[v] \iff \forall s \preceq t$, not $(\mathcal{M} \models d_s(\phi)[v])$
6. $\mathcal{M} \models d_s(d_t(\phi))[v] \iff \mathcal{M} \models d_t(\phi)[v]$
7. $\mathcal{M} \models d_t(\forall x_i \phi)[v] \iff \forall a \in \mathcal{M}, \mathcal{M} \models d_t(\phi)[v(i/a)]$
8. $\mathcal{M} \models d_t(\exists x_i \phi)[v] \iff \exists a \in \mathcal{M}, \mathcal{M} \models d_t(\phi)[v(i/a)]$

Global satisfaction is achieved if and only if $\mathcal{M} \models d_t(\Psi)[v]$ holds true for all active nodes $t \in T$, proving that the transferred local ontology is mutually compatible across the homomorphic security perimeters.